SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and *Big Data Analytics* **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

I THIRU KUMARAN S / 192521149 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thiru Kumaran S

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Evolution of cloud Computing

Hardware evolution	internet Software evolution	Distributed System	service Delivery
includes	includes	includes	includes
* main frame * client - server * virtualized * modern data centers	* static website * dynamic web apps * web 2-0 * APPs and web services	* multiple servers * Load balancing * fault tolerance * Scalable architecture	* Iaa, (infra) * Paa, (platform) * Saa S (Software) * xaa (anything as a Services

Modern cloud platform,
(flexible + Scalable + on-demand).

DEFINITION:

Cloud Computing is the delivary of Computing resource storage, software and over the Internet on demand

is

Cloud Computing fundamentals

has

CHARACTERISTICS:

* Elasticity
* Scability
* Resource pooling
* on-demand access
* measured service

has

LIMITATIONS:

* Security and privacy risk
* Dependence on internet
* Less Control over data
* potential downtime
* Data transfer cost

provides

BENEFITS:

* Cost Saving
* High availability
* flexibility
* Better resource utilisation
* Global access

cloud computing

Key component
* vm
* Hyperviso..
* Abstraction
* Hardware resources

visualitation
↓
allows
↓
* multiple vm to run on a
* single physical
* hardware
*
↓
Benefits
↓
* Better resource
* Isolation & Security
* flexibility portability
* faster provising

web Series
↓
provides
↓
* Standard way for
* application to Commun
* Catim over the internet
* (using APIS)
↓
Benefits
↓
* interoperability system
* loose Coupling
* Language
* Independence

Key Component
* API
* SOAP
* REST
* provides

Together enable
↓
cloud deployement
* Scalability

* on demand
* Resource
* Elasticity

* flexible
* high
* cost

Cloud Service Models

Iaas

* provides virtual servers, network,
* user manages os and applications.
* eg: AWS. EC2, Azure VM

Paas

* provides development platform
* user manages applications and data
* eg: Google App Engine

Saas

* provides ready-to-use software
* user just uses it
* eg: Gmail, Google drive

Xaas

* Extension of cloud
* includes backups, security, DB, etc.
* eg: Baas. Daas SECaas